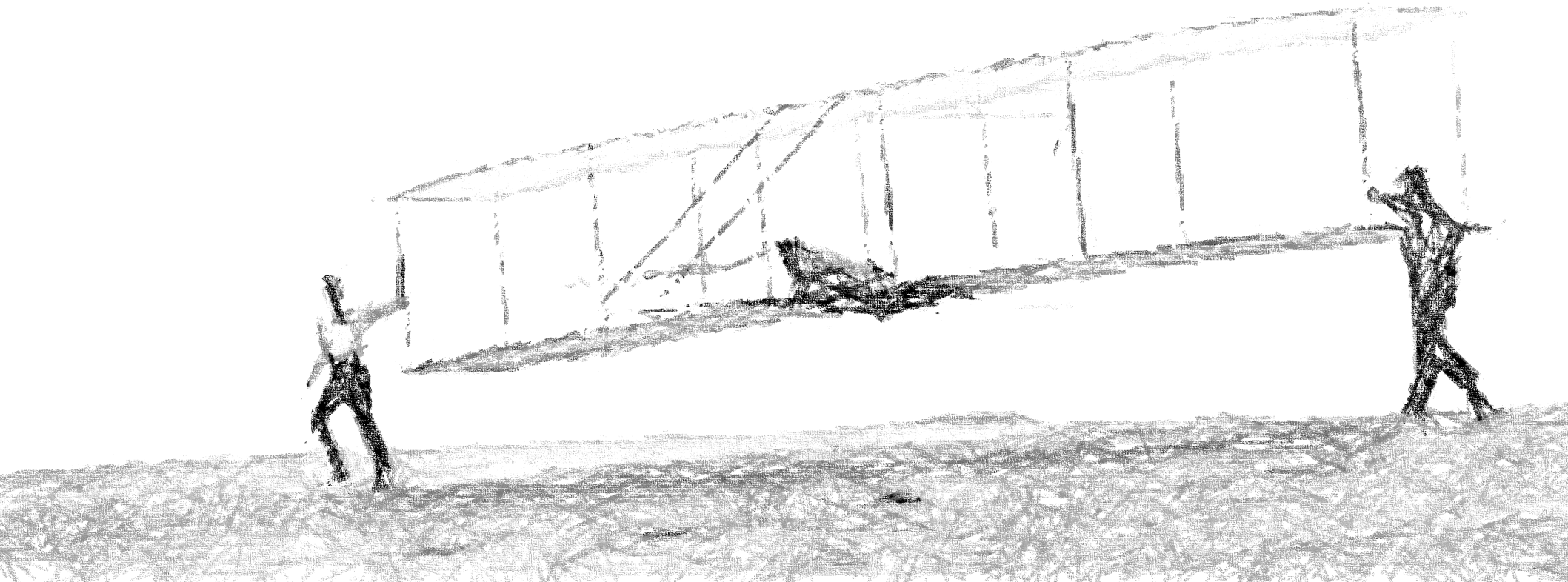




Mechanics applied to Aerospace Engineering

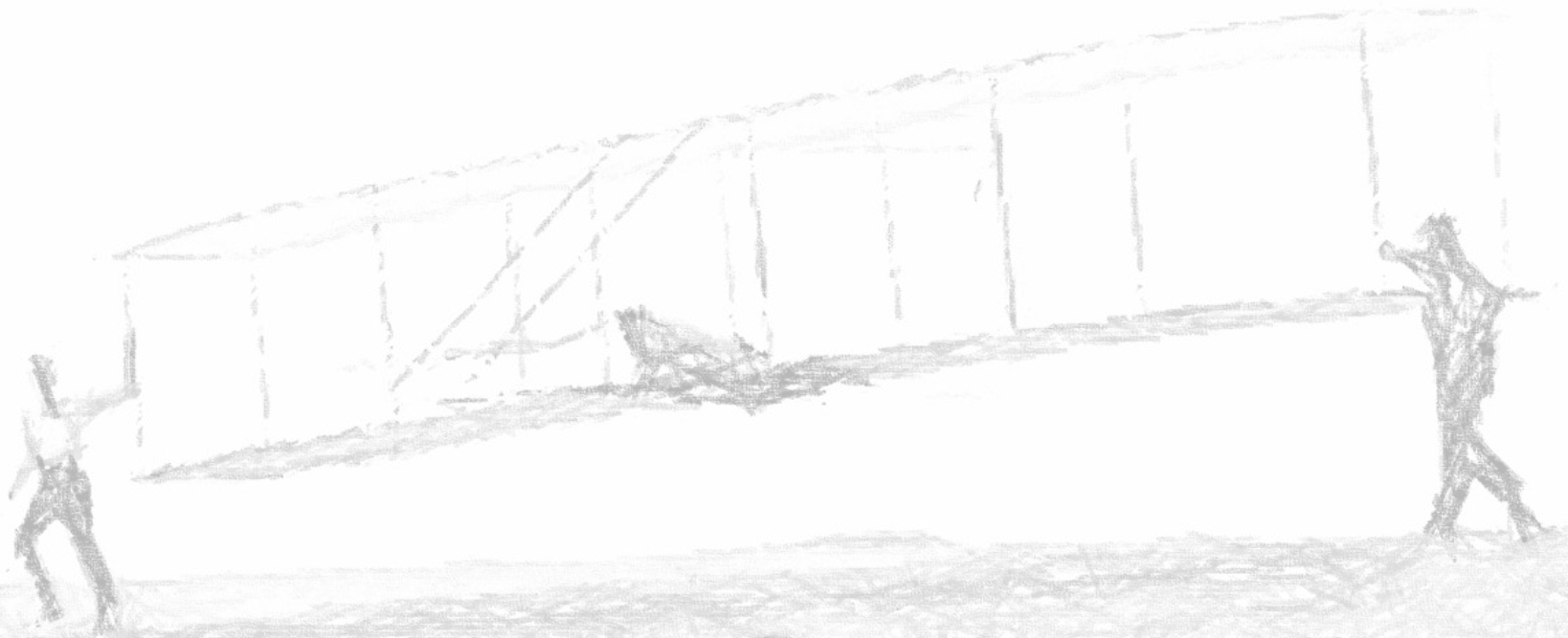
Course 251-14165 – BSc Aerospace Engineering

4. Angular momentum and torque



Contents

- Angular momentum
- Torque
- Differential equation
- Conservation of angular momentum
- Central force problems
- Kepler's problem

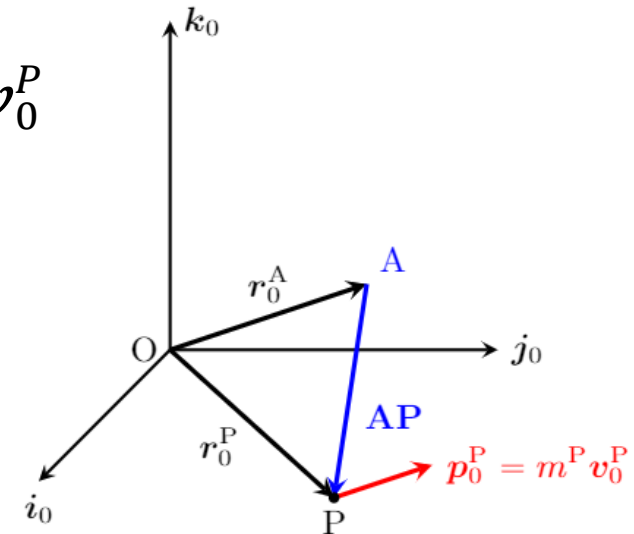


Angular momentum and torque

- The angular momentum of a point particle P about a point A, computed in reference frame S_0 , is the vector quantity:

$$\mathbf{H}_{A0}^P = \mathbf{AP} \times \mathbf{p}_0^P = (\mathbf{r}_0^P - \mathbf{r}_0^A) \times m^P \mathbf{v}_0^P$$

- It is the *moment* of $\mathbf{p}_0^P$ about A
- Its value depends on point A and the reference frame used to compute $\mathbf{v}_0^P$
- Units: ML^2/T (Kg m²/s)



- The torque of a force $\mathbf{F}$ about a point A is:

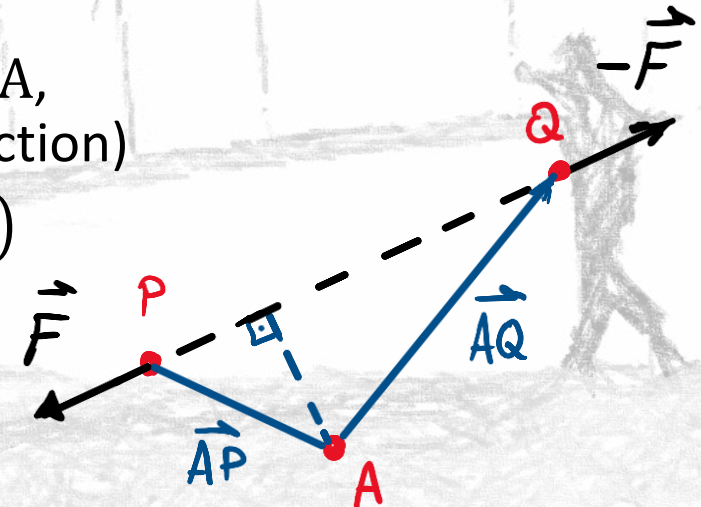
$$\mathbf{M}_A = \mathbf{AP} \times \mathbf{F} = (\mathbf{r}_0^P - \mathbf{r}_0^A) \times \mathbf{F}$$

- It is also a vector (Units: ML^2/T^2 (N m))
- It is the moment of $\mathbf{F}$ about A

- If a particle Q causes a torque $\mathbf{M}_A$ on P about A, then P causes a torque $-\mathbf{M}_A$ on Q (action-reaction)

$$\mathbf{AP} \times \mathbf{F}^P = -\mathbf{AQ} \times \mathbf{F}^Q = -\mathbf{AQ} \times (-\mathbf{F}^P)$$

- Torques can be classified into *internal* and *external*, relative to our system of analysis



Differential equation of H_{A0}^P

- Differentiating with respect to time in $H_{A0}^P = (\mathbf{r}_0^P - \mathbf{r}_0^A) \times m^P \mathbf{v}_0^P$:

$$\left. \frac{dH_{A0}^P}{dt} \right|_0 = \underbrace{(\mathbf{r}_0^P - \mathbf{r}_0^A)}_{\vec{AP}} \times \underbrace{m^P \mathbf{a}_0^P}_{\vec{F}} + (\mathbf{v}_0^P - \mathbf{v}_0^A) \times m^P \mathbf{v}_0^P$$

$\vec{v}_0^P \times \vec{v}_0^P = \vec{0}$

$$\left. \frac{dH_{A0}^P}{dt} \right|_0 = \underbrace{AP \times F}_{\vec{M}_A} - \mathbf{v}_0^A \times m^P \mathbf{v}_0^P$$

$$\left. \frac{dH_{A0}^P}{dt} \right|_0 = \mathbf{M}_A - \mathbf{v}_0^A \times m^P \mathbf{v}_0^P$$

- If $\mathbf{v}_0^A = \mathbf{0}$ (i.e., if A is a fixed point in S_0), the equation simplifies to:

$$\left. \frac{dH_{A0}^P}{dt} \right|_0 = \mathbf{M}_A$$

- This differential equation gives the time evolution of H_{A0}^P

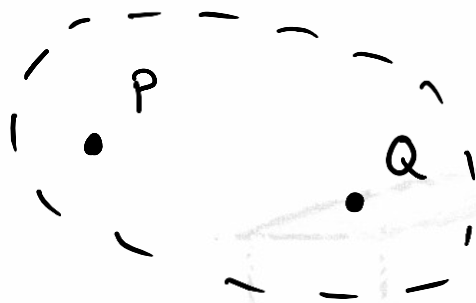
It is a derived equation from Newton's second law. It does not add new information

Conservation of angular momentum

- The angular momentum about a fixed point A of a particle P is conserved if the net torque about A acting on P is zero:

$$\mathbf{M}_A = \mathbf{0} \Rightarrow \left. \frac{d\mathbf{H}_{A0}^P}{dt} \right|_0 = 0 \Rightarrow \mathbf{H}_{A0}^P = \mathbf{const}$$

- The *total* angular momentum about a fixed point A of a system of analysis is constant if the net external torque about A acting on the system is zero
 - *An isolated system satisfies this condition*

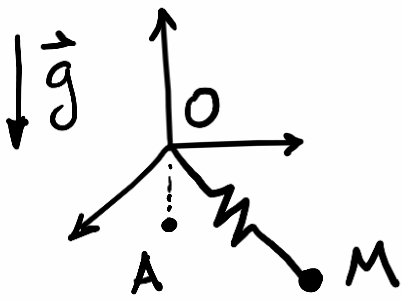


No external torque
 $\Rightarrow$ conservation of
total angular momentum

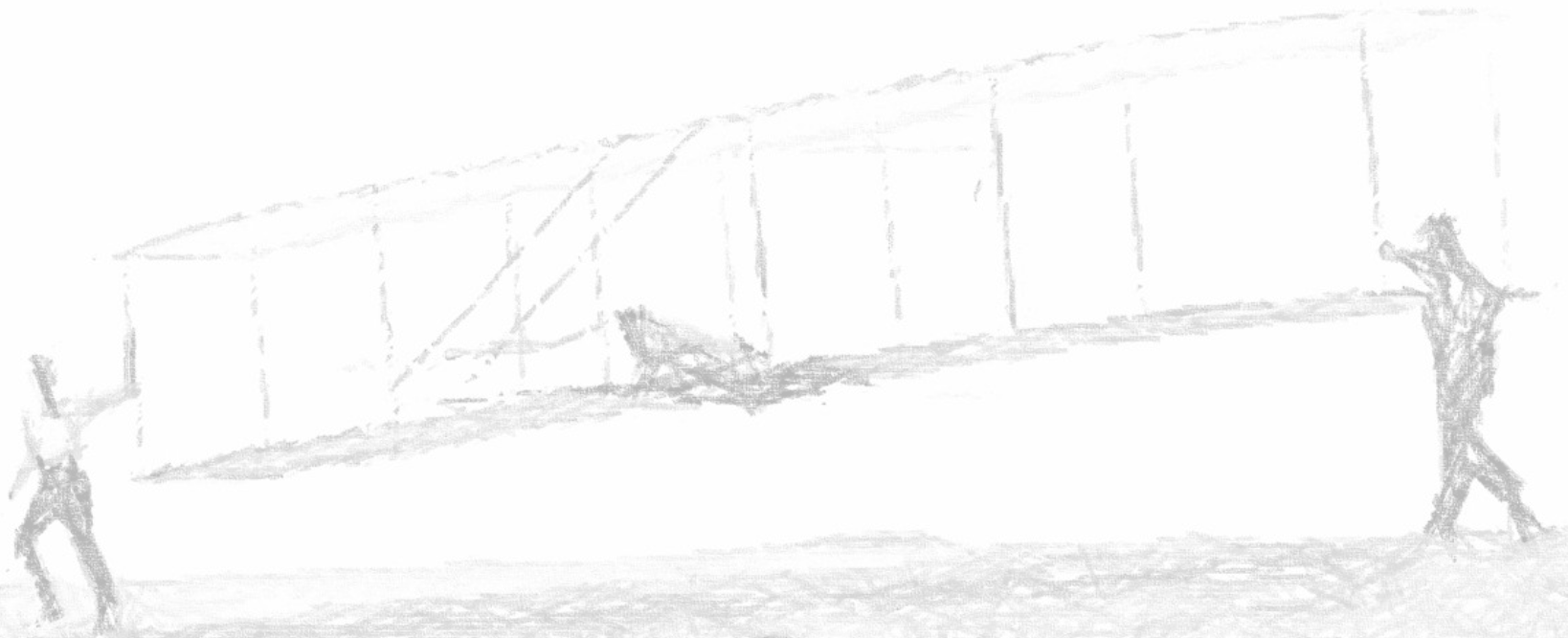
The conservation of angular momentum (vector) is equivalent to three scalar conservation laws.

Example

A force attracts a heavy particle M of mass m to a point O . The attractive force is proportional to the mass of the particle and to the distance OM . The proportionality coefficient is k^2 . The coordinate system $Oxyz$ has the origin at the centre of attraction O and the axis Oz is vertically oriented. Initially, the coordinates of M are $(x_0, y_0, z_0) = (\sqrt{3}g/k^2, 0, 0)$ and its initial velocity is $(\dot{x}_0, \dot{y}_0, \dot{z}_0) = (0, 2g/k, 0)$.

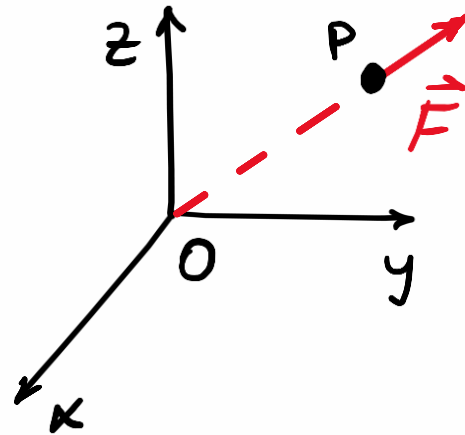


$$\begin{cases} \ddot{x} = -k^2 x \\ \ddot{y} = -k^2 y \\ \ddot{z} = -g - k^2 z \end{cases}$$



Central force problems

- We say that a problem is a *central force problem* with respect to reference frame S_0 if the line of action of the net force $\mathbf{F}$ acting on P always passes through the origin O

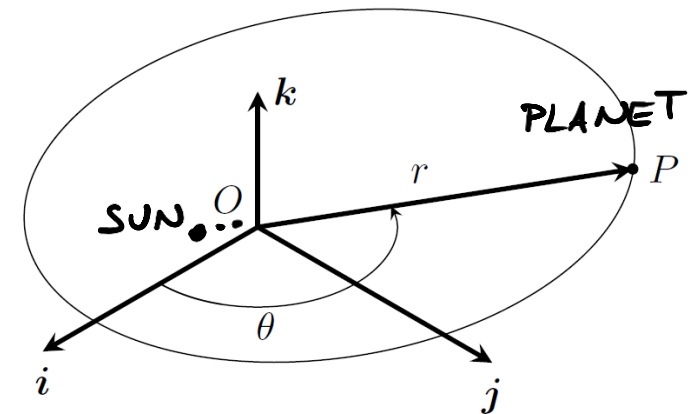


$$\vec{F} = F \vec{e}_r$$

- Central force problems have a number of interesting properties:
- The angular momentum vector $\mathbf{H}_{O0}^P$ is constant. This gives three scalar conservation laws
 - Since $\mathbf{r}_0^P, \mathbf{v}_0^P \perp \mathbf{H}_{O0}^P$, the motion of P occurs in a plane perpendicular to $\mathbf{H}_{O0}^P$
 - These problems are easily analyzed in polar coordinates in the plane of motion
 - Since $H_{O0}^P = mR^2\dot{\theta} = \text{const}$, the angular rate increases when R decreases, and vice-versa

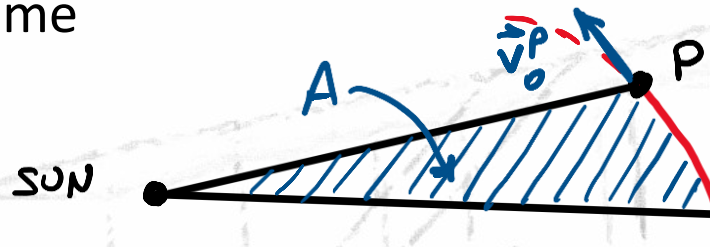
Kepler's problem

- Kepler's problem, or the two body problem, concerns the motion of two point particles under their mutual gravitational attraction
 - It is a great model for many problems in orbital mechanics (motion of a satellite around the Earth; motion of the Earth around the Sun)



Kepler's laws are based on empirical observations

1. The orbits of the planets are ellipses, with the Sun at one of the foci.
2. The line joining a planet to the Sun sweeps out equal areas in equal intervals of time



$$\frac{dA}{dt} = \text{const}$$

3. The square of the period of a planet is proportional to the cube of the major axis of its elliptical orbit ($\tau^2 \propto a^3$)

However, Kepler gave no explanation as to why this is so.

Kepler's problem

➤ Let us try to prove Kepler's laws from the Newtonian framework:

- Newton's law of motion
- Newton's law of universal gravitation ($F_g = Gm_P m_S / d^2$)

➤ In an inertial reference frame S_0 :

$$m^P \frac{d\mathbf{v}_0^P}{dt} \bigg|_0 = - \frac{Gm^P m^S}{|\mathbf{r}_0^P - \mathbf{r}_0^S|^2} \frac{(\mathbf{r}_0^P - \mathbf{r}_0^S)}{|\mathbf{r}_0^P - \mathbf{r}_0^S|}$$

$$m^S \frac{d\mathbf{v}_0^S}{dt} \bigg|_0 = + \frac{Gm^P m^S}{|\mathbf{r}_0^P - \mathbf{r}_0^S|^2} \frac{(\mathbf{r}_0^P - \mathbf{r}_0^S)}{|\mathbf{r}_0^P - \mathbf{r}_0^S|}$$

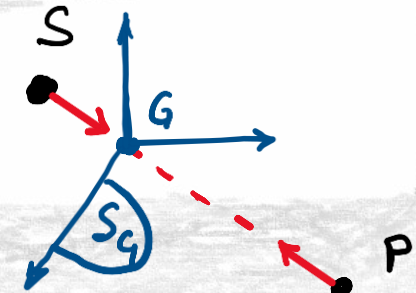
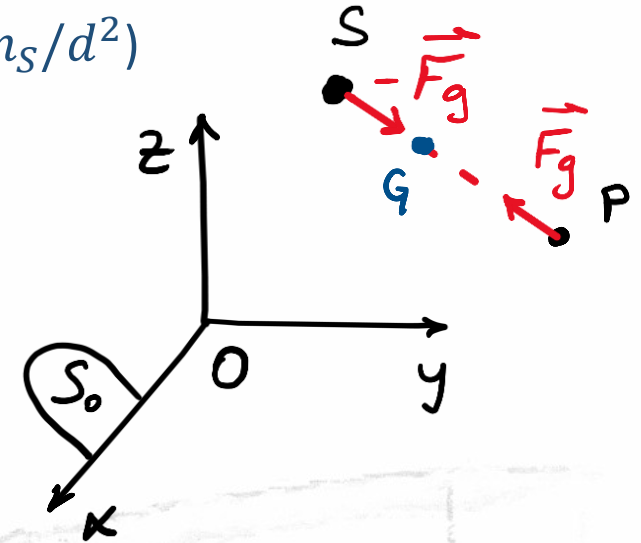
➤ Center of mass equation (i.e., the barycenter G):

$$m^S \frac{d\mathbf{v}_0^S}{dt} \bigg|_0 + m^P \frac{d\mathbf{v}_0^P}{dt} \bigg|_0 = (m^S + m^P) \frac{d^2 \mathbf{r}_0^G}{dt^2} \bigg|_0 = \mathbf{0}$$

- The center of mass G is a non-accelerated point

- A reference frame S_G parallel to S_0 but with origin at G is also inertial

- Since they are parallel, $\frac{d}{dt} \bigg|_0 = \frac{d}{dt} \bigg|_G = \frac{d}{dt}$



Kepler's problem

- Motion of P relative to S . Let $\mathbf{r} = \mathbf{r}_0^P - \mathbf{r}_0^S$:

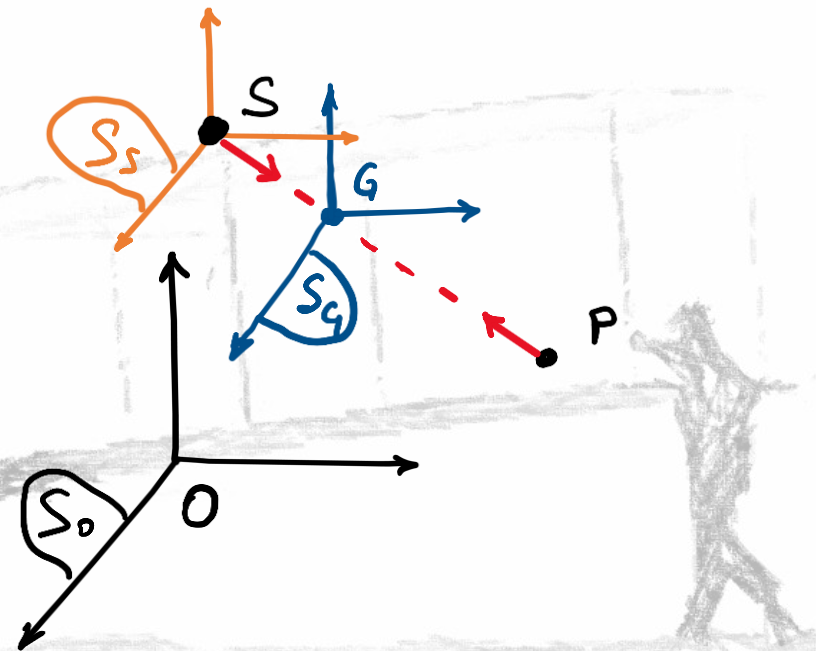
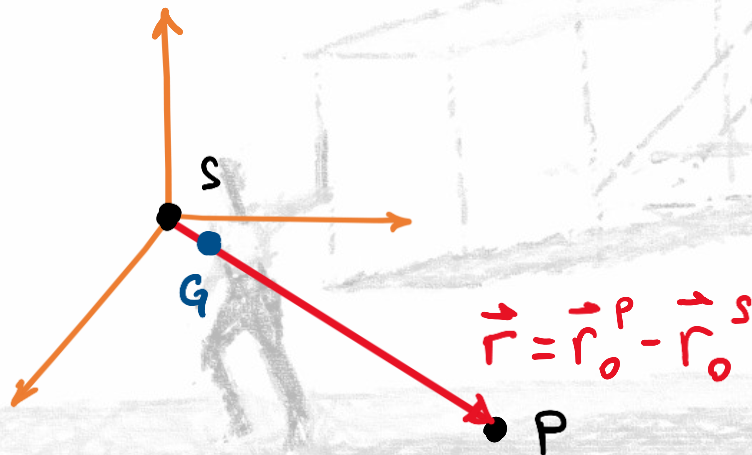
$$\frac{d\mathbf{v}_0^P}{dt} = - \frac{Gm^S}{|\mathbf{r}_0^P - \mathbf{r}_0^S|^2} \frac{(\mathbf{r}_0^P - \mathbf{r}_0^S)}{|\mathbf{r}_0^P - \mathbf{r}_0^S|}$$

$$\frac{d\mathbf{v}_0^S}{dt} = + \frac{Gm^P}{|\mathbf{r}_0^P - \mathbf{r}_0^S|^2} \frac{(\mathbf{r}_0^P - \mathbf{r}_0^S)}{|\mathbf{r}_0^P - \mathbf{r}_0^S|}$$

$$\frac{d^2\mathbf{r}}{dt^2} = - \frac{G(m_S + m_P)}{r^2} \frac{\mathbf{r}}{r}$$

We call $\mu = G(m^S + m^P)$

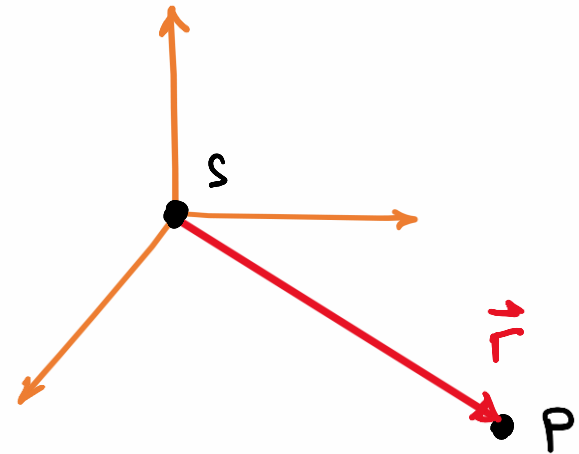
- We introduce S_S , parallel to S_0 and S_G , but with origin at S . Since S is an accelerated point, it is non-inertial
- Since $m^S \gg m^P$, $\mu \simeq Gm^S$, $G \sim S$, and non-inertial effects are negligible



Kepler's problem

- Final equation:

$$\frac{d^2 \mathbf{r}}{dt^2} = -\frac{\mu}{r^3} \mathbf{r}$$



- This is a central force problem!
 - Angular momentum is conserved
 - Motion is planar
 - Let's use polar coordinates in the plane of motion
- The mass of P does not appear in the problem!
 - The motion of P does not depend on its mass
 - Let's use mass-specific variables (mass-specific angular momentum, energy...)
- Additionally, it is a conservative problem!
 - Gravity has (mass-)specific potential energy $V_0 = -\mu/r$

Kinematics and dynamics

- In polar (cylindrical) coordinates (r, θ) :

$$\mathbf{r} = r \mathbf{e}_r$$

$$\mathbf{v} = \dot{r} \mathbf{e}_r + r \dot{\theta} \mathbf{e}_\theta$$

$$\mathbf{a} = (\ddot{r} - r \dot{\theta}^2) \mathbf{e}_r + (r \ddot{\theta} + 2 \dot{r} \dot{\theta}) \mathbf{e}_\theta$$

- Equations of motion:

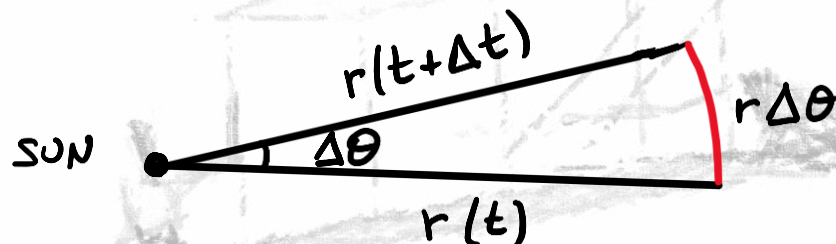
$$\frac{d^2 \mathbf{r}}{dt^2} = -\frac{\mu}{r^2} \mathbf{e}_r \quad \begin{array}{ll} \mathbf{e}_r: & \ddot{r} - r \dot{\theta}^2 = -\mu/r^2 \\ \mathbf{e}_\theta: & r \ddot{\theta} + 2 \dot{r} \dot{\theta} = 0 \end{array}$$

- Multiplying the $\mathbf{e}_\theta$ equation by r we can integrate:

$$r^2 \dot{\theta} = h = \text{const}$$

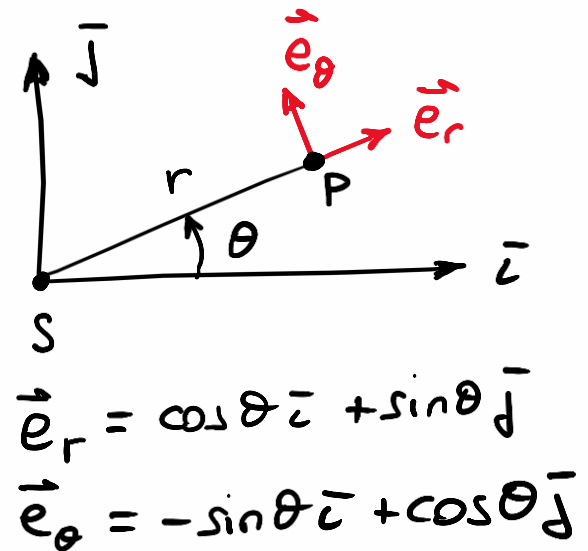
- This is the magnitude of the mass-specific angular momentum vector!

- Proving second law of Kepler:



$$\Delta A = \frac{r \cdot r \Delta\theta}{2}$$

$$\frac{dA}{dt} = \frac{1}{2} r^2 \frac{d\theta}{dt} = \frac{1}{2} h = \text{const}$$



Proving 1st Kepler's law

- The equation along $\mathbf{e}_r$ can also be integrated:

$$\frac{d^2 r}{dt^2} - r\dot{\theta}^2 = -\frac{\mu}{r^2}$$

Remember:
 $\dot{\theta} = h/r^2$

$$\frac{dr}{dt} = \frac{d\theta}{dt} \frac{dr}{d\theta} = \dot{\theta} \frac{dr}{d\theta} = \frac{h}{r^2} \frac{dr}{d\theta} = -h \frac{d}{d\theta} \left(\frac{1}{r} \right)$$

$$\frac{d^2 r}{dt^2} = \dot{\theta} \frac{d}{d\theta} \left[-h \frac{d}{d\theta} \left(\frac{1}{r} \right) \right] = -\frac{h^2}{r^2} \frac{d^2}{d\theta^2} \left(\frac{1}{r} \right)$$

$$-\frac{h^2}{r^2} \frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) - \frac{h^2}{r^3} = -\frac{\mu}{r^2} \Rightarrow \frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) + \left(\frac{1}{r} \right) = \frac{\mu}{h^2}$$

- This is analogous to the equation of a forced harmonic oscillator!
- The solution gives the equation of a conic section (ellipse, parabola, hyperbola) in polar coordinates:

$$r = \frac{h^2/\mu}{1 + e \cos \theta}$$

Trajectories are conic sections

➤ Trajectory equation:

$$r = \frac{h^2/\mu}{1 + e \cos \theta}$$

- Conic section with parameter $p = h^2/\mu$ and eccentricity e
- Useful relations for conic sections: $p = a(1 - e^2)$, $b = a\sqrt{1 - e^2}$ where a is the semi-major axis, b the semi-minor axis
- In the ellipse case, the pericenter is at $\theta = 0$, the apocenter at $\theta = \pi$
 $r_p = a(1 - e)$; $r_a = a(1 + e)$
- Not only ellipses are possible (e.g. planets). Hyperbolas, parabolas too (e.g. comets)

	a	e	} 1 st Kepler law cases
circle	$a > 0$	$e = 0$	
ellipse	$a > 0$	$0 < e < 1$	
parabola	$a = \infty$	$e = 1$	
hyperbola	$a < 0$	$e > 1$	

- The trajectory equation gives us $r = r(\theta)$, but we do not know $\theta = \theta(t)$.

Orbital period

- The area of an ellipse of semi-axes a, b is $A = \pi ab$
- We know the area rate is

$$\frac{dA}{dt} = \frac{h}{2}$$

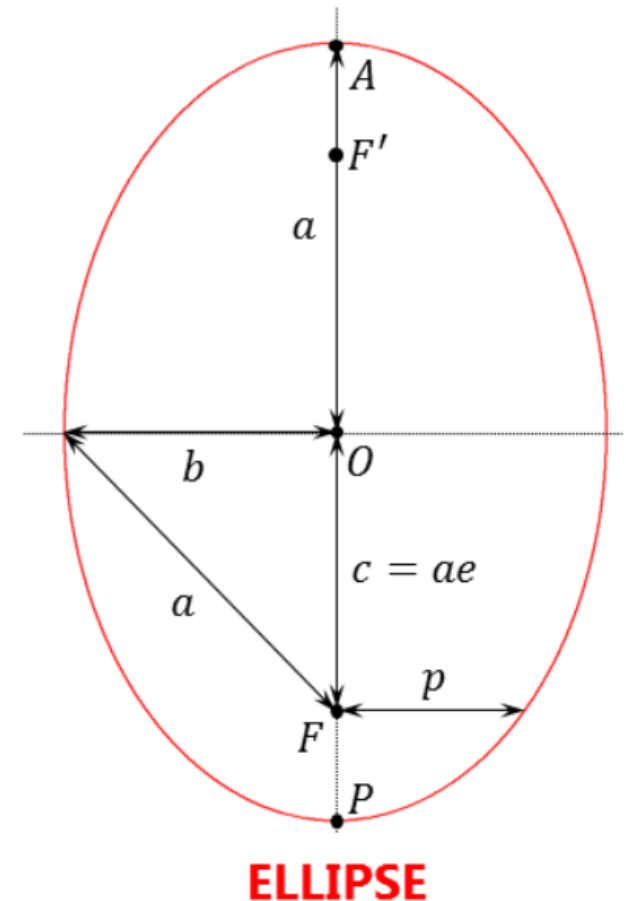
- Integrating in a full orbit:

$$\frac{h}{2} \int dt = \int dA$$

$$\tau = \frac{2\pi ab}{h} = 2\pi \sqrt{\frac{a^3}{\mu}}$$

- This proves Kepler's third law:

$$\tau^2 = \frac{4\pi^2}{\mu} a^3$$



Vis-viva equation and fundamental velocities

- The mass-specific mechanical energy ξ is conserved. Particularizing at the pericenter and the apocenter we find the value of this constant:

$$\xi = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}$$

- Mechanical energy is negative for ellipses, zero for parabolas, positive for hyperbolas
- If the orbit is circular ($r = a$), the velocity is:

$$v_c = \sqrt{\frac{\mu}{a}}$$

- This is the *circular velocity*, which decreases with the distance to S

- The minimum velocity to escape the system and reach infinity is:

$$v_e = \sqrt{\frac{2\mu}{r}}$$

- This is the *escape velocity*, which also decreases with distance to S

- Particularizing at the pericenter and apocenter we find:

$$v_p = \sqrt{\frac{\mu(1+e)}{a(1-e)}}; \quad v_a = \sqrt{\frac{\mu(1-e)}{a(1+e)}}$$

Thank you!

